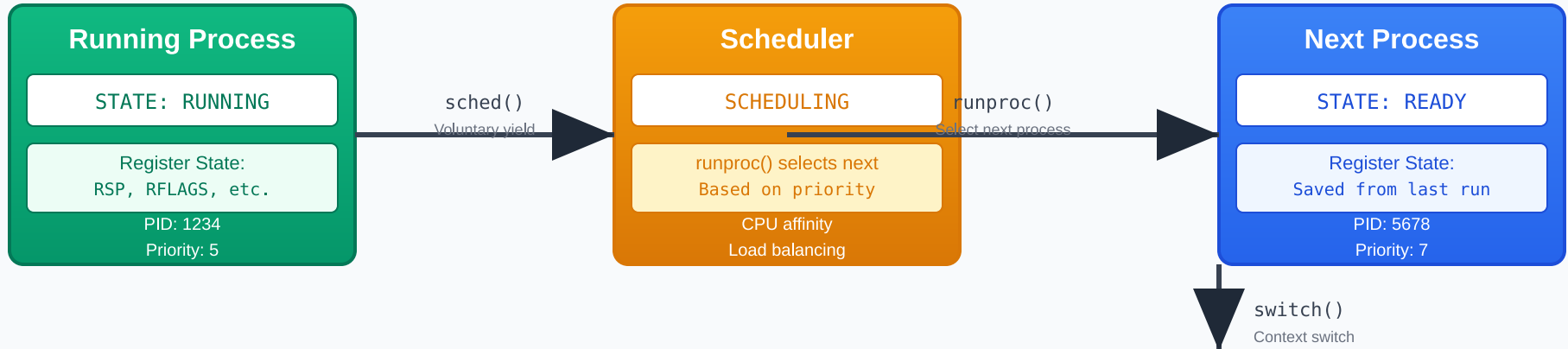


Context Switch Flow

From Running Process to Scheduler to Next Process



Context Switch Steps

1. Save State

Save current process registers to Proc

2. Select Next

`runproc()` chooses highest priority

3. Load State

Load next process registers from Proc

4. Update State

Switch CR3, Mach
Begin execution

Register State Saved/Restored

- General purpose: RAX, RBX, RCX, RDX, RSI, RDI, RBP, RSP
- Instruction pointer: RIP, Flags: RFLAGS
- Segment registers: CS, SS, DS, ES, FS, GS
- Control registers: CR3 (page table root)
- Debug registers: DR0-DR7
- FPU/SSE state: XMM, YMM registers

Context Switch Overhead

- ~1-2 microseconds on modern x86_64 CPUs
- TLB flush required when switching CR3